

TCS349: Responsible and Explainable AI

Unit 3, Lecture 2

Job Augmentation: Human-AI Collaboration & Redesigned Work

How AI can extend human capability rather than replace it - and what it takes to redesign work so people and AI genuinely complement each other.

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Job Augmentation



Human-AI Collaboration



Redesign of Work



New Skills

1. Job Augmentation - What It Means

Job augmentation means AI takes on specific sub-tasks within a role while a human continues to perform the tasks that require judgement, context or interpersonal skill.

Unlike full automation, augmentation keeps the human in the workflow as an active decision-maker, not a passive supervisor.

Augmentation typically targets the most time-consuming or error-prone parts of a role, freeing capacity for higher-value work.

The same AI capability can be deployed as either automation or augmentation, depending entirely on how the workflow is designed around it.

2. Human-AI Collaboration Models

Sequential collaboration: AI completes an initial pass - a draft, a prediction, a summary - and a human reviews or finishes it.

Parallel collaboration: AI and human work on different parts of a task simultaneously, then combine results.

AI-as-assistant: a human directs the work and calls on AI as a tool for specific sub-tasks, on demand.

The right model depends on how much judgement a task requires and how costly an AI error would be if left unchecked.

3. Scenario: Augmentation in Healthcare

A radiologist uses an AI system that pre-screens scans and highlights regions likely to contain an abnormality.

The radiologist still makes the final diagnosis, using the AI's highlights as a starting point rather than a verdict.

This sequential model lets the radiologist review more scans in the same time, while catching cases that manual review alone might miss.

The design choice to keep a human as final decision-maker directly reflects the human-oversight principle from Unit II.

4. Scenario: Augmentation in Customer Service

A support agent uses an AI assistant that suggests responses and surfaces relevant account information in real time during a call.

The agent chooses whether to use, edit or discard each suggestion based on the specific customer's situation.

This AI-as-assistant model reduces time spent searching for information, letting the agent focus on the conversation itself.

Response quality and customer satisfaction still depend on the agent's judgement, even though the AI shapes what is readily available to them.

5. Redesigning Work Around AI

Simply adding an AI tool to an unchanged workflow rarely produces the full benefit of augmentation.

Effective redesign reallocates a worker's time from the automated sub-tasks toward the judgement-based tasks AI cannot perform.

Roles, performance metrics and training often need to be updated alongside the technology itself, not left as they were.

Workers who are not involved in redesigning their own workflow are more likely to resist or under-use a new AI tool.

6. New Skills Needed for Human-AI Collaboration

AI literacy: understanding what an AI tool can and cannot reliably do, including its likely failure modes.

Critical evaluation: knowing when to trust an AI suggestion and when to override it, rather than accepting outputs by default.

Prompting and tool direction: framing a task clearly enough for an AI system to produce a useful starting point.

These skills increasingly matter across many occupations, not only in technical or AI-specialist roles.

7. Risks of Poorly Designed Augmentation

Automation bias: workers may over-trust AI suggestions, reducing the quality of the human review step the design assumed would catch errors.

Skill erosion: heavy reliance on AI for a sub-task can gradually weaken a worker's own ability to perform that task unaided.

Increased workload: augmentation can add oversight burden without removing enough of the original task, leaving workers busier rather than more effective.

These risks reinforce why augmentation must be designed deliberately, using the accountability and oversight principles from Unit II.